

Mwaslaty API Flow Support Review

Backend support assessment for the six product flows and key edge cases

Executive Summary

The current backend exposes one route-planning endpoint, **POST /api/plan**. It is a strong first slice for route search and comparison, but it is not yet a full product API for navigation, AI trip planning, saved account data, payments, tickets, scanning, or operator workflows.

Existing support includes route search, route cards, estimated fares, transfers, modes, leg instructions, and basic OTP-related errors. Most remaining flows require new backend contracts and domain-specific error codes.

Current API Surface

- POST /api/plan validates from.lat, from.lng, to.lat, to.lng, date, and time.
- Optional preferences include modes and optimizeFor.
- Successful responses include planId, source, optimizedFor, from, to, itineraries, legs, durationMinutes, totalFare, transfers, summary, route names, and basic instructions.
- Current error codes: VALIDATION_ERROR, OTP_EMPTY_PLAN, OTP_SERVICE_UNAVAILABLE, OTP_GRAPHQL_ERROR, and INTERNAL_SERVER_ERROR.

Six Flow Support Matrix

Flow	Support	Evidence and gaps
Flow 1: Plan A to B	Partial	POST /api/plan supports coordinates, date/time, modes, route search, route results, and no-route handling. Missing geocoding/autocomplete and AI prompt parsing.
Flow 2: Compare route options	Partial	Route cards include duration, estimated fare, transfers, modes, summary, and legs. Sorting exists for quickest, cheapest, most_comfortable, and reliable. Missing formal comfort score and route detail endpoint.
Flow 3: Follow step-by-step	Mostly missing	Legs include basic instructions, times, distances, and stops. Missing navigation sessions, GPS updates, deviation detection, recalculation, offline sync, arrival confirmation, and feedback.
Flow 4: Save places/routes	Client-only	The diagram stores data in IndexedDB, so no backend is required for local-only saving. Server sync endpoints are missing if account-based saved screens are needed.
Flow 5: AI Trip Planner	Missing	No endpoint for AI prompt analysis, budget/duration parsing, attractions/restaurants, multi-day schedules, cost breakdown, save, or share.
Flow 6: Booking and payment	Missing	No booking, quote, payment, ticket generation, QR payload, validation status, ticket scan, refund, or operator authorization API.

Missing Endpoints

- GET /api/places/search for destination suggestions and autocomplete.
- POST /api/ai/route-intent for natural-language route requests.
- GET /api/plans/:planId or GET /api/routes/:itineraryId for route detail reload.
- POST /api/navigation/sessions, PATCH /api/navigation/sessions/:id/location, and POST /api/navigation/sessions/:id/recalculate.
- POST /api/feedback.

- Saved data endpoints if account sync is required: `/api/saved/places` and `/api/saved/routes`.
- `POST /api/ai/trips` for the AI Trip Planner.
- `POST /api/bookings/quote`, `POST /api/bookings`, `POST /api/payments/confirm`.
- `GET /api/tickets/:ticketId` and `POST /api/tickets/:ticketId/scan`.
- Authentication and operator authorization endpoints.

Missing Fields

- Route geometry or polyline for map rendering.
- Stop IDs, GTFS IDs, platform names, and coordinates per leg.
- Detailed step list per leg beyond a single instruction string.
- `fare.status`, `fare.source`, and `fareConfidence`.
- `comfortScore`, `crowding`, `disruption alerts`, `realtime delay`, and `accessibility metadata`.
- Ticket eligibility, `bookingId`, `paymentId`, `ticketId`, `qrPayload`, `expiresAt`, and `validationStatus`.
- Offline/cache metadata such as `cacheable`, `validUntil`, and `lastSyncedAt`.

Missing Error Codes

- `GEOCODING_NO_MATCH`, `AI_PARSE_FAILED`, `ROUTE_OUTSIDE_SERVICE_HOURS`, `ROUTE_PARTIAL_ONLY`, `FARE_UNKNOWN`.
- `NAVIGATION_OFFLINE_NO_CACHE`, `NAVIGATION_LOCATION_UNAVAILABLE`, `ROUTE_DEVIATION_RECALCULATION_FAILED`.
- `PAYMENT_FAILED`, `TICKET_EXPIRED`, `TICKET_ALREADY_USED`, `TICKET_WRONG_LEG`, `TICKET_SCAN_OFFLINE_PENDING`.
- `AUTH_REQUIRED`, `FORBIDDEN`, `OPERATOR_UNAUTHORIZED`.

Edge Case Response Matrix

Edge case	HTTP	Code	Expected API response
No results	404	OTP_EMPTY_PLAN	Return error details with from, to, date, and time. Current API already does this for empty itineraries.
Partial routes	200 or 409	ROUTE_PARTIAL_ONLY	Return status partial with available legs, unavailable legs, and whether booking/navigation can continue.
Fare unknown	200	FARE_UNKNOWN warning	Return fare amount null, currency EGP, status unknown, and fareConfidence 0 rather than silently using 0.
Off-hours	422	ROUTE_OUTSIDE_SERVICE_HOURS	Return service window, timezone Africa/Cairo, and next available service time.
Signal loss mid-trip	200/503	NAVIGATION_LOCATION_UNAVAILABLE	Keep session in offline_tracking when cached route exists. If server action is required, return recoverable unavailable error.
Ticket scanned twice	409	TICKET_ALREADY_USED	Return original scannedAt, stationId, legId, and ticket status used.
Wrong-leg scan	409	TICKET_WRONG_LEG	Return expected leg/route and scanned leg/route. Do not validate the ticket.
Expired ticket	410	TICKET_EXPIRED	Return expiresAt and optional rebook action metadata.
Scan while offline	202 or 503	TICKET_SCAN_OFFLINE_PENDING	Accept signed offline QR scans as pending sync. Otherwise require online validation.
Operator not authorized	403	OPERATOR_UNAUTHORIZED	Reject scan request without leaking ticket details. Include required operator scope.

Recommended Next API Slice

The most useful next backend increment is to harden route planning before starting payment or AI workflows.

- Add place search/autocomplete and route detail reload.
- Add route geometry, stop IDs, fare status, and comfort/reliability metadata.
- Separate no-route, off-hours, partial-route, and fare-unknown responses.
- Then add navigation sessions and feedback.
- Treat booking, payment, tickets, and operator scanning as a later bounded module with its own security model.